Nicolas Desjardins

Lead Systems Engineer Turned Applied AI Architect

Eight years owning production web platforms, now building deterministic AI systems: schema-enforced extraction, hybrid retrieval, agent state machines, and LLMOps.

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Greater Montreal Area, QC, Canada
Local, hybrid, or remote US and Canada

8+	30+	$90k+	40%+	200k+	56
years in production	properties migrated, zero downtime	licensing eliminated	infrastructure cost cut	lines of TypeScript (NextBlock)	MCP tools for coding agents

CORE STACK

Applied AI

Python · FastAPI · Pydantic v2 · LangGraph · LangChain · Anthropic SDK · OpenAI SDK · OpenRouter · MCP · Vercel AI SDK · Ollama

Retrieval & data

PostgreSQL · pgvector · tsvector / BM25 · Cohere Rerank · Redis · Supabase · MySQL

Web

Next.js 16 · React 19 · TypeScript · Tailwind CSS v4 · shadcn/ui · Laravel · PHP · Node.js · Tiptap · Nx

Infrastructure

Docker · Linux · cPanel / WHM · VPS · Vercel · Cloudflare Workers · GitHub Actions · Tailscale

LLMOps & testing

Langfuse · DeepEval · Ragas · Pytest · Vitest

LANGUAGES

French, full professional proficiency
English, full professional proficiency

EDUCATION

Computer Science, General Program
Concordia University, Montreal
2016 to 2018

DEC in Computer Science
John Abbott College,
Sainte-Anne-de-Bellevue
2009 to 2012

MILITARY SERVICE

RCEME Weapons Technician, Canadian Armed Forces
2013 to 2016, CFB Valcartier, QC

PROFILE

Lead web developer and systems engineer with 8+ years of production ownership across 30+ digital properties, and the creator of NextBlock CMS, an open-source AI-native CMS for Next.js 16 and Supabase (200k+ lines of TypeScript, 8 npm packages, an MCP server with 56 tools). I bring what production AI actually needs: typed schemas, fallbacks that are tested, telemetry, and evaluation gates. Available for Senior Applied AI, AI Solutions, and Forward Deployed Engineer roles.

EXPERIENCE

Creator & Principal Engineer

NextBlock CMS (open source) · Oct 2025 to present · Greater Montreal Area, QC

- Designed and built an AI-native CMS as an Nx monorepo: Next.js 16 Server Components, Supabase Postgres with row-level security, a Tiptap block editor, a commerce module, a CLI, and the release pipeline for 8 npm packages (v0.21.3).
- Shipped Cortex AI: an MCP server over Streamable HTTP with scoped tokens and 56 tools, confirmation-gated mutations, typed Zod tool calling with SSE streaming, OpenRouter routing with free-model fallback, and AES-256-GCM key storage.
- Built a zero-environment-variable deployment: Vercel's Supabase integration provisions the database and forward-only migrations apply during the first build; one command self-hosts the full stack on Docker.
- 100/100 Lighthouse on the production build, 146 Vitest files, 191 CLI releases, and 11.5k downloads in the 30 days to 7 October 2026.

Lead Web Developer & Applied AI Integration

New Roots Herbal Inc. · Nov 2018 to present · Vaudreuil-Dorion, QC

- Rebuilt newrootsherbal.com on Laravel 13 with a custom CMS and e-commerce store written from scratch, a Google Maps store locator, a real-time inventory system, and 18 campaign microsites.
- Deployed an internal multi-model generative AI asset pipeline (Claude, OpenAI Codex, Gemini) that cut catalog asset turnaround from weeks to 40 minutes per SKU, a 10x gain in throughput.
- Consolidated relational databases and deployed hybrid retrieval (lexical plus vector) with strict Pydantic validation, retiring legacy vendors and saving $90,000+ in licensing.
- Directed infrastructure across 30+ properties with zero-downtime migrations and a 40%+ cloud cost reduction; telemetry and attribution funnels captured $300,000+ in incremental revenue in one 30-day campaign.

EARLIER EXPERIENCE

Technical Founder

PcSquad · 2012 to 2018 · Les Coteaux, QC

- Founded and ran a regional IT solutions and web infrastructure consultancy: scoping, custom software, deployment, systems administration, networking, and hardware diagnostics for small and mid-sized businesses.
- Owned the financial, legal, and marketing side, including SEO and paid acquisition; resolved hundreds of client IT problems.

Founder

Webman Solutions · 2016 to 2018 · Greater Montreal Area, QC

- Web studio: built client websites and handled their early SEO and SEM.

Webmaster & Technician

Zakiry Productions · 2005 to 2018 · Pincourt, QC

- Website, CRM, and sound systems for an entertainment production house; webmaster and computer technician.

SELECTED PROJECTS

Measured numbers come from saved benchmark records, recorded runs, and public registries.

Invoice Extraction Service

Production live, open source

An asynchronous FastAPI microservice that turns messy, unstructured documents into a guaranteed, schema-validated invoice record. Resilience comes from routing across models, not vendors: a free primary model cascades to a second provider on rate limits, timeouts, or schema exceptions, at zero marginal cost.

`21 / 21 fully correct, primary model` `10 failure classes, offline` `0 network calls in tests` `$0.00 per request`

nickdesjardins.github.io/Invoice-Extraction-Service github.com/nickdesjardins/Invoice-Extraction-Service

Context-Aware Hybrid Compliance Retrieval Engine

Live demo, open source

Hybrid RAG over SOPs and regulatory manuals: pgvector and full-text search fused by Reciprocal Rank Fusion in one PostgreSQL statement, a cross-encoder keeps 3 passages, and every quote is verified against the stored text before it is shown. Runs at zero cost on free models.

`6 / 6 answers with verified quotes` `1 / 1 out-of-corpus declined` `441 MB peak, 512 MB host` `$0.00 per question`

nickdesjardins.github.io/hybrid-knowledge-rag github.com/nickdesjardins/hybrid-knowledge-rag

NextBlock CMS

Flagship product, v0.21.3, AGPL core

The open-source, full-stack AI-native CMS for Next.js 16, Supabase, and Tailwind CSS. Prompt builders hand you code to host and patch. WordPress hands you plugins to babysit. NextBlock turns typed content blocks in PostgreSQL into a website a whole team can run: a Notion-style editor for writers, React Server Components for the public site, and an MCP server so coding agents can build pages over the protocol they already speak.

- Content engine: every block is a JSONB row in PostgreSQL, custom block types are data, pages publish with no rebuild, Live Draft Mode and revisions, EN/FR content, SEO graded as you type.
- Data and security: three roles enforced by a database trigger, RLS kept Supabase Advisor clean, 21 forward-only migrations applied at build time, encrypted bring-your-own-key storage.
- Cortex AI: MCP server listed in the official registry (56 tools), page-aware agent with typed tools, OpenRouter routing with fallback, no token markup.
- Commerce Pro: Stripe and Freemius checkout, multi-currency with rounding modes, scheduled sales, shipping zones, taxes, inventory.

nextblock.dev GitHub: github.com/nextblock-cms/nextblock npm: www.npmjs.com/package/create-nextblock

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Milestones and web work · webman.dev · nick.desjardins@live.com

SCHEDULED MILESTONE ARCHITECTURES

Phases 3 and 4 of a 120-day Applied AI plan whose first two phases have shipped (page 2). Figures are design targets, not measurements.

Autonomous Multi-Agent Operations Pipeline

Delivery: Days 61 to 90

A cyclic LangGraph workflow that coordinates SQL inventory queries, channel-specific copy synthesis, and regulatory compliance auditing. Validation failures loop back with structured revision notes. High-risk mutations pause for a human before anything leaves the system.

Targets: 100% of high-risk mutations gated by a human checkpoint · Full state replay from any checkpoint · 2-tier model cascade by task complexity

Production LLMOps Gateway & Evaluation CI/CD Suite

Delivery: Days 91 to 120

An enterprise reverse proxy in front of every model call. Semantically similar queries resolve from a Redis vector cache in milliseconds, every transaction is traced, and a CI pipeline runs regression evaluations against golden datasets so a bad prompt change never reaches main.

Targets: < 15 ms cache hit response time · 0.96 cosine threshold for cache resolution · 5% hallucination ceiling that blocks a PR · ~65% inference cost reduction from caching

PRODUCTION WEB WORK

Employer work is credited to New Roots Herbal Inc.; the rest is independent.

newrootsherbal.com (employer)
Laravel 13 · PHP · MySQL · Blade · JavaScript · Google Maps API · Linux VPS

Full platform revamp on Laravel 13: custom CMS and e-commerce store, store locator, inventory system.

18 campaign microsites (employer)
Next.js · Laravel

Next.js microsites and Laravel landing pages for product lines and health topics.

Subdomains: vitamins, aging, women, digestion, omegas, stress, probiotics, mushrooms, magnesium, immune, oils (.newrootsherbal.com)

Own domains: coughandcold.ca, prostateperform.ca

Platform pages: newrootsherbal.com/cardiac-health, /vitamin-d, /ukraine, /pain-management, /essential-oils

MusiqueSD (client)
Next.js · React

Bilingual marketing site for a Montreal recording studio, DJ and event business, with a quote-request flow.

Ryūgū: Ten Nights to the Dragon Palace
Next.js · React

Chaptered travel companion for an eleven-day voyage around Japan and Jeju.

Thieves' Cant Translator
Next.js · React · v0

D&D 5e rogue's companion: English or French to Thieves' Cant, with dictionary and symbols.

Experiments
Next.js · React · Vercel

Civalgo Punch, a real-time worker check-in system; an interactive React Hooks guide.

pcsquad.ca (own business, 2012)
HTML · CSS · JavaScript · PHP

The bilingual site of my first company, a computer repair and website shop I owned and ran from 2012 to 2018.

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Technical arsenal · webman.dev · nick.desjardins@live.com

TECHNICAL ARSENAL

195 tools across 12 domains, from the 2012 consultancy years to the current Applied AI work. Full detail with usage context at webman.dev.

Applied AI & LLM orchestration 13

FastAPI · Pydantic v2 · LangGraph · LangChain · Tiktoken · Anthropic SDK · OpenAI SDK · Google GenAI SDK · Groq · OpenRouter · Ollama · Model Context Protocol · Vercel AI SDK

Frontier & open models 9

Claude · ChatGPT / GPT · Gemini · DeepSeek · Qwen · Perplexity · Mistral · Llama · Nemotron

Retrieval, vectors & data 11

PostgreSQL · pgvector (HNSW / IVFFlat) · tsvector / BM25 · Cohere Rerank · ChromaDB · Qdrant · Redis · MySQL · phpMyAdmin · SQLite · MongoDB

Languages 21

Python · TypeScript · PHP · JavaScript · SQL · HTML5 · CSS3 · Sass / SCSS · Bash · PowerShell · Batch · C++ · C# · Java · Dart · Visual Basic · XML · JSON · YAML · Blade · Markdown

Frameworks & modern web 23

Next.js (App Router) · React · Tailwind CSS · shadcn/ui · Supabase · Tiptap · Laravel · Node.js · Nx Monorepo · Stripe / Freemius · Zod · Three.js / React Three Fiber · Vite · Angular · Bootstrap · jQuery · WordPress · WooCommerce · Drupal · SuiteCRM · Gatsby · Scully · Flutter

Infrastructure & systems 31

Docker · Docker Compose · Linux · Debian · Ubuntu · Windows · Android · Red Hat · CentOS · Fedora · cPanel / WHM · VPS Hosting · Apache · Nginx · WAMP / LAMP · GitHub Actions · Vercel · Cloudflare (Workers, R2) · Render · DigitalOcean · Google Cloud Platform · Tailscale · Netlify · Google App Engine · Firebase · VirtualBox · Hyper-V · VMware Workstation / vSphere · WinSCP · PuTTY · FileZilla

LLMOps & evaluation 5

Langfuse Tracing · DeepEval · Ragas · Pytest · Vitest

Developer tooling 21

VS Code · Git · GitHub · Git Bash · Claude Code · Cursor · OpenAI Codex · Jules · npm · Postman · Chrome DevTools · Lighthouse · Jira · Asana · Notepad++ · GitHub Desktop · Visual Studio · Android Studio · Eclipse · Sublime Text · Internet Explorer

Design & media 19

Figma · Adobe Photoshop · Adobe Illustrator · Adobe InDesign · Adobe Premiere Pro · Adobe After Effects · Adobe Acrobat · GIMP · Blender · OBS Studio · Audacity · Studio One · Melodyne · Google Fonts · Adobe Fonts · Font Awesome · SketchUp · Adobe Dreamweaver · Adobe Flash

Marketing & analytics 11

Google Analytics · Google Tag Manager · Google Search Console · SEO / SEM · Google Ads · Meta Ads · Google Maps Platform · Google Business Profile · Google Trends · PayPal · Google AdSense

Productivity & collaboration 15

Microsoft 365 · Microsoft Excel · Microsoft Teams · Microsoft SharePoint · Microsoft Visio · Microsoft Access · Google Workspace · Google Sheets · Slack · Discord · Zoom · LibreOffice · OpenOffice · Evernote · Skype

IT support & security 16

Malwarebytes · ESET NOD32 · Kaspersky · Norton · McAfee · Avira · Microsoft Security Essentials · CCleaner · AdwCleaner · Ultimate Boot CD · DxDiag · Chrome Remote Desktop · WatchGuard · Cisco Packet Tracer · 7-Zip · VLC